



IEEE AESS
Sustainability
Hackathon

Aether Avionics

Track 1

LEO Cube Satellite Power
Management System

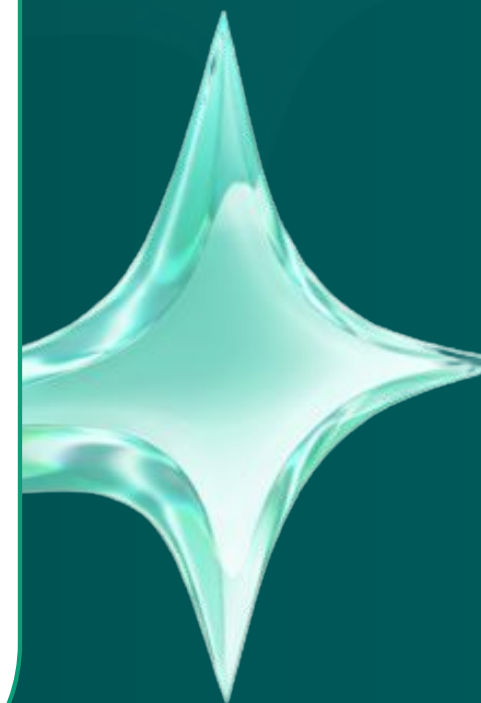


Problem Statement

The Dual-Threat of Orbital Extremes

CubeSat environmental missions face a critical dual-threat to their longevity: **energy starvation and thermal overload**.

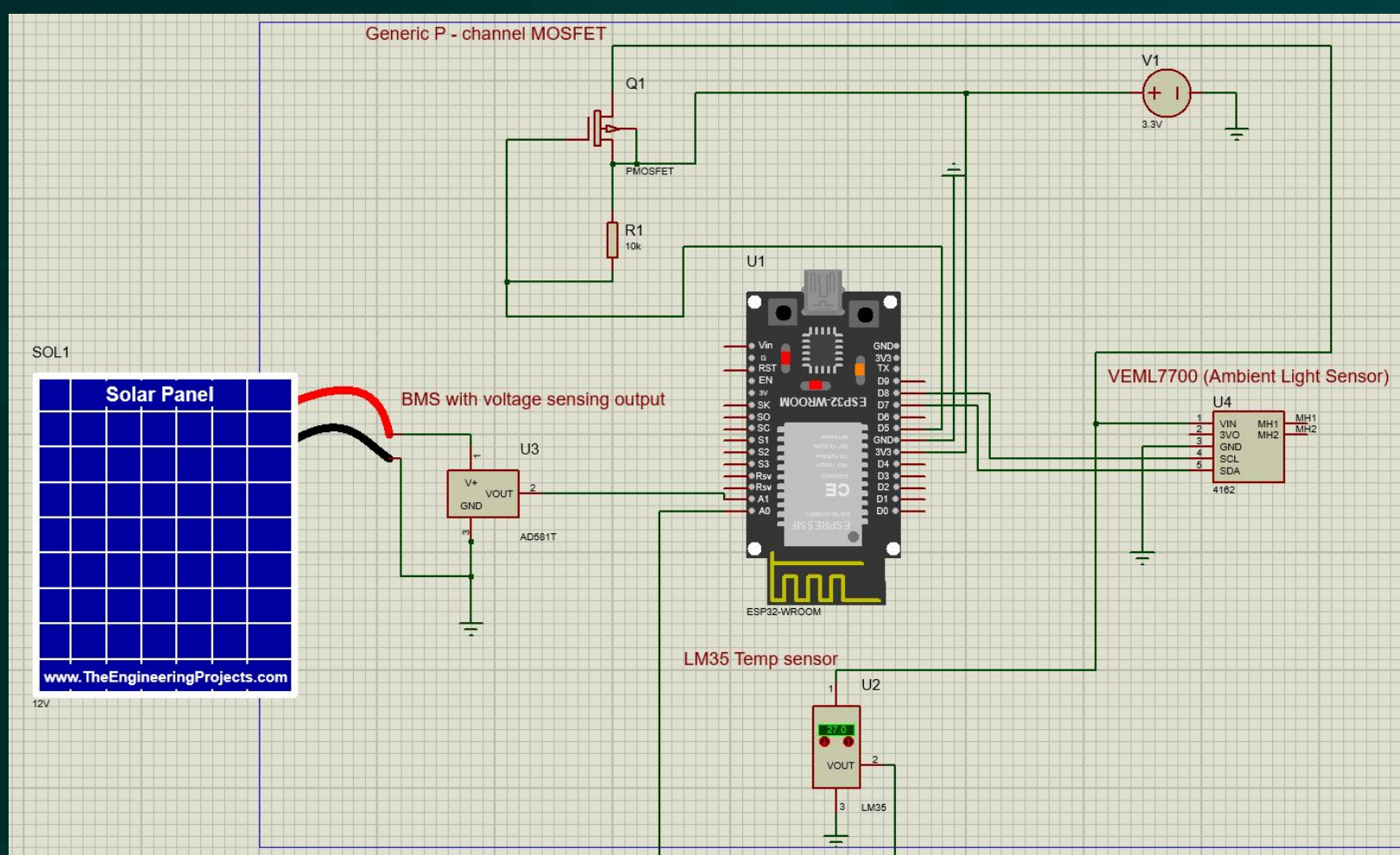
During high-albedo operations, prolonged solar eclipses cause satellites to overuse battery life by blindly running idle systems and sensors 24/7 in the dark. Conversely, when exposed to direct orbital radiation, intense heat can push microcontrollers to a catastrophic 120°C, **risking permanent hardware meltdown**. Because current architectures lack the environmental awareness to dynamically power down during these extreme light and heat cycles, they suffer severely compromised mission lifespans and hardware reliability.



Proposed Solution

- We have designed an *Autonomous Power Management System* designed for small satellites. It acts as a digital survival instinct for the CubeSat, ensuring it never drains its batteries to failure by syncing its workload with the solar cycle.
- Depending on the position of the CubeSat in orbit, the required equipment is turned on and off for power optimization.
- We highlight the difference between active and passive management as well as dual-purpose hardware.

System Architecture & Technical Design



- Hardware components: ESP32, P-channel MOSFET, VEML7700 Ambient Light Sensor, LM35 Temperature sensor.
- Data flow: Sunlit → ESP32 active, MOSFET open, VEML7700 sampling → solar panel signal drops (eclipse) → MOSFET cuts sensor rail, ESP32 enters 60s deep-sleep cycle → wakes every 60s, checks panel signal + LM35 temp → if still eclipse or temp $\geq 120^{\circ}\text{C}$, sleeps again → solar generation resumes + temp safe → MOSFET reopens rail, back to active sampling.

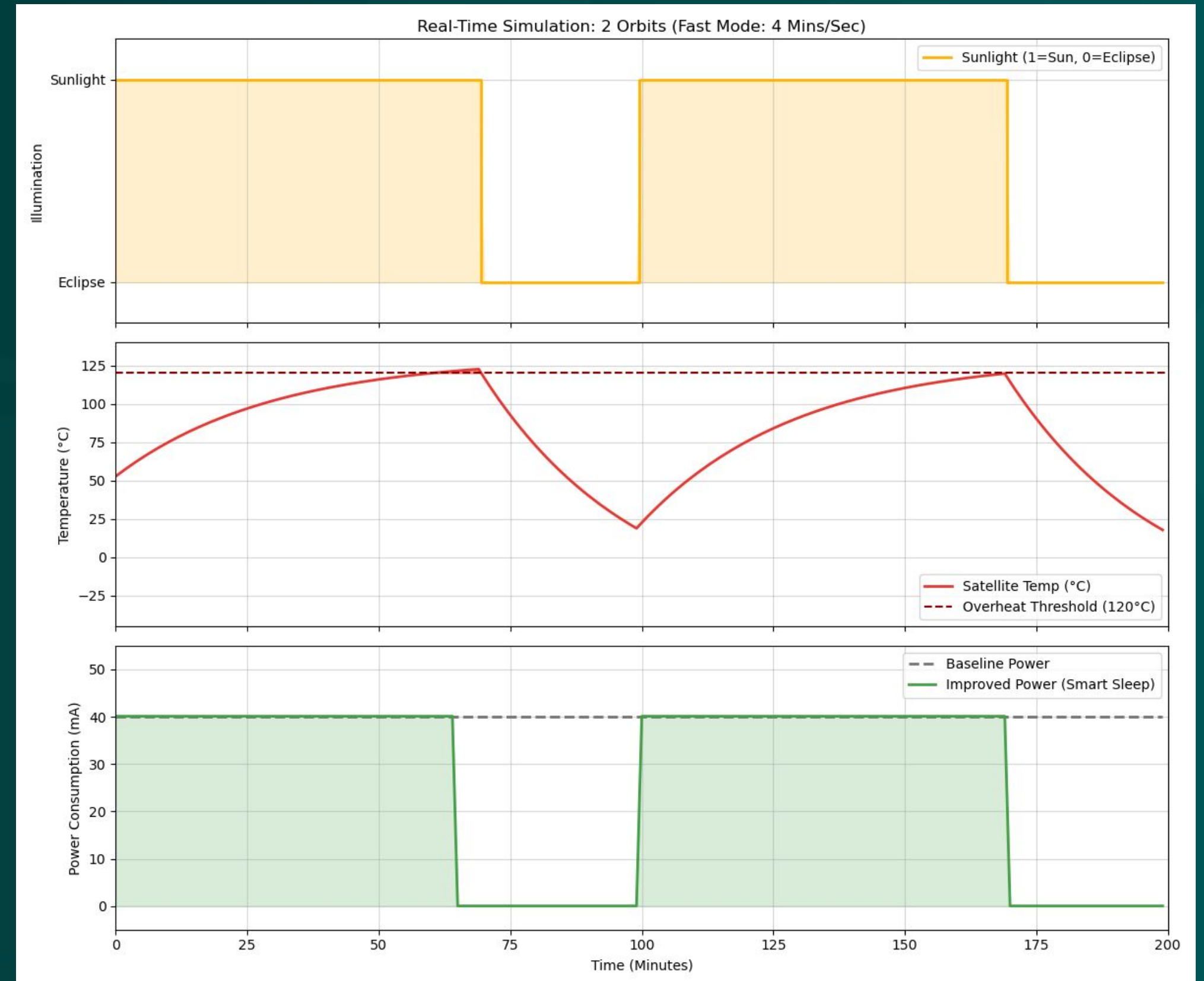
Technical Implementation

- **Technologies Used:**
IoT and embedded systems
- **Software Development:**
VScode, Proteus, Spice, C++, Python
- **Main Communication Protocol:**
I2C



Results / Prototype / Demo :

Screenshot of graph results from our simulation



Results / Prototype / Demo :

Mission Summary & Savings			
Metric	Baseline System (Always On)	Improved System (Smart Sleep)	Total Saved
Total Energy Per Orbit	0.220049 Wh	0.154053 Wh	0.065996 Wh
Energy per Day (14.4 Orbits)	3.168706 Wh	2.218363 Wh	0.950343 Wh / Day
Battery Life Extension	1.0x (Baseline)	1.42x	+42% Longer Life
Overall Power Reduction	0%	30.00%	30.00% Less Power

Screenshot of power savings

Impact

- A satellite's most dangerous moment is when it loses power and becomes uncontrolled debris. Our system ensures satellites remain powered and controllable, contributing to a cleaner, safer orbital environment for future generations.
- We are making space missions more sustainable by giving small satellites the autonomous power intelligence needed to survive longer, reduce orbital debris, and provide the granular climate data required to track Earth's melting ice caps.

SDG Alignment

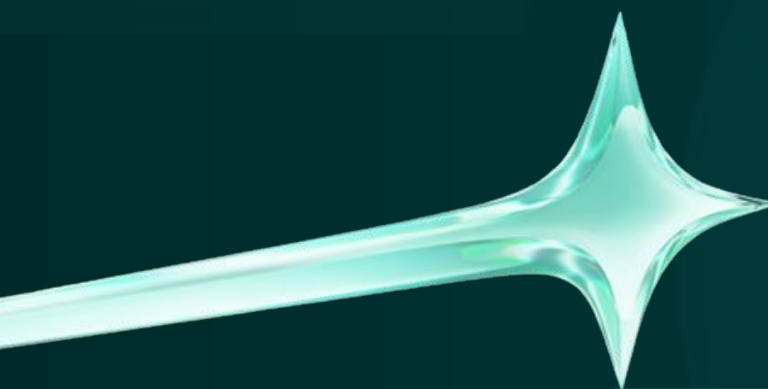
- We are targeting SDGs 9, 12, 13, and 17.
- When it comes to space projects, these are of the most basic yet overlooked values.
- By increasing these satellites' lifespans we ensure more sustainable use of materials, smarter innovation and through less pollution we directly aid climate action.



Scalability & Future Work

- To make this space hardened, specifically for radiation mitigation we can use triple modular regulation to ensure no errors were caused by cosmic rays or kept at a minimum at least.
- As an example in real life deployment, we can employ a meshnetwork logic. If one CubeSat in a fleet enters an eclipse, it could “hand off” its tasks to a nearby satellite thats close to the sun. This ensures climate monitoring of a specific Earth region is never interrupted, even as individual satellites pass into darkness.
- For future improvements, instead of a simple on/off threshold we intend to implement Maximum power point tracking. This uses software to constantly adjust the electrical load on the solar panels to extract the maximum wattage possible, even when the sun is hitting the panels at an awkward angle.





Challenges & Limitations

Challenges:

Our biggest challenge was designing a thermal protection system that didn't trigger a false positive. We had to model the thermal inertia of the satellite so it wouldn't react to instant spikes.

Limitations:

The transition from active sampling to sleep mode depends on the solar panel signal dropping, introducing a short lag before the MOSFET responds. The 60-second polling interval also means sunlight restoration is detected with up to a 60-second delay

Conclusion

Our autonomous power-gating logic transforms "disposable" hardware into resilient scientific assets by ensuring every joule of energy is preserved for critical climate insights.